

The University of Azad Jammu and Kashmir

Name: Zarnab Fatima

Roll NO: 2024-SE-14

Semester:5th

Course Title: Digital Image Processing

Course Code: CS-3112

Submitted To: Engr. Fahad Nisar

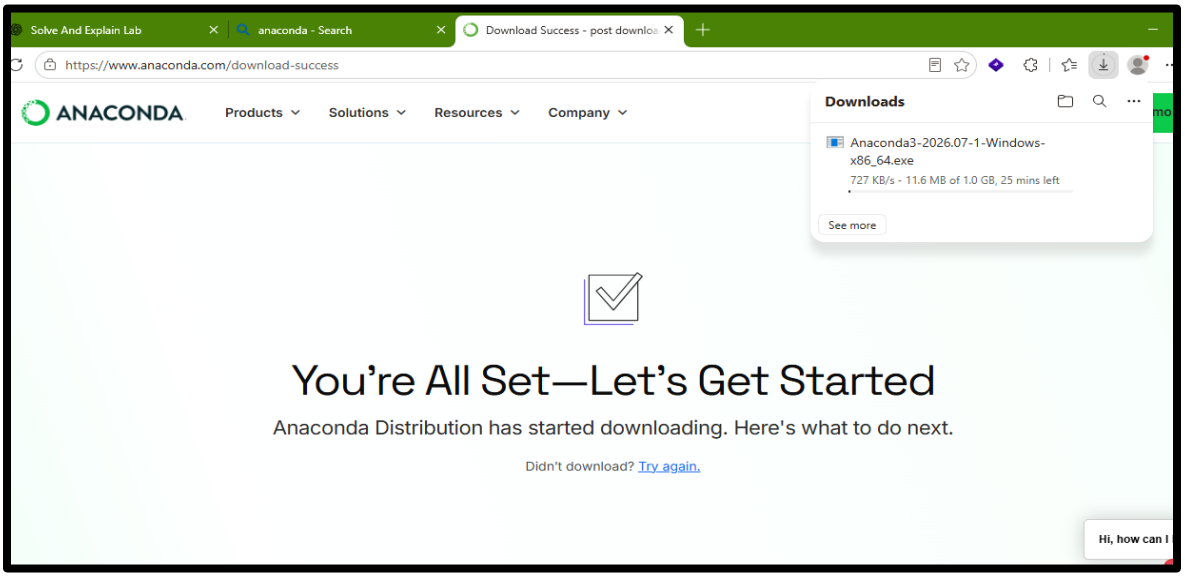
Department of Software Engineering

LAB 01

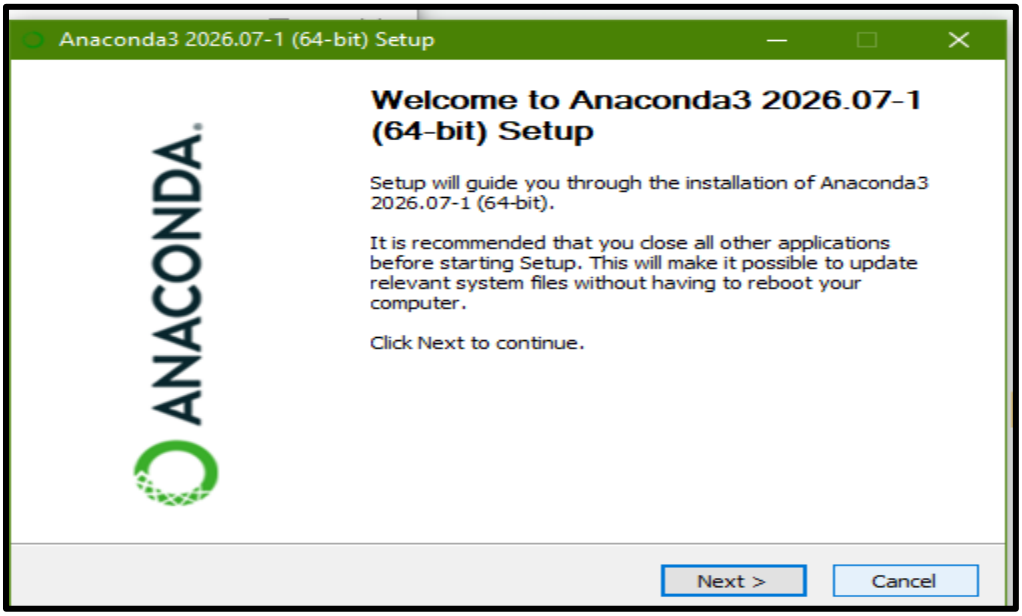
➢ Part 1

Task 1: Install Anaconda

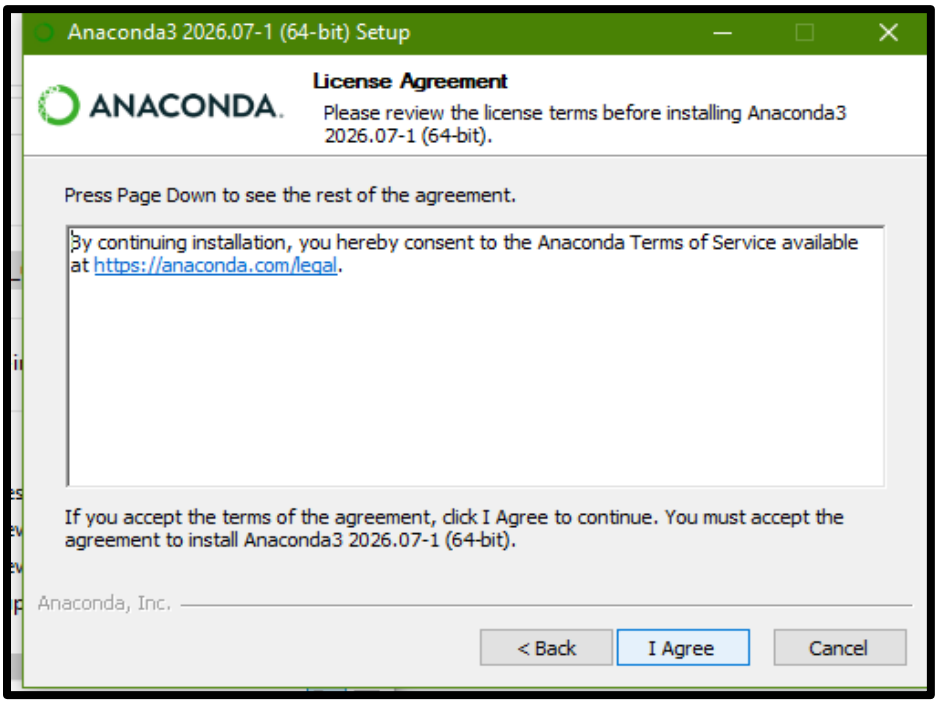
Step 1:



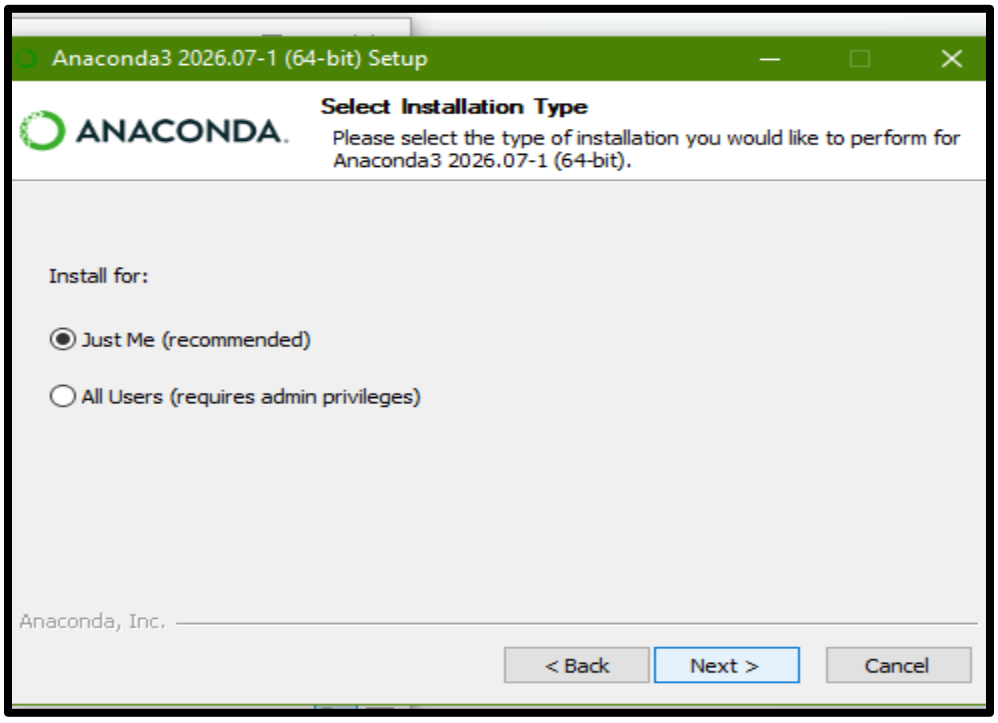
Step 2:



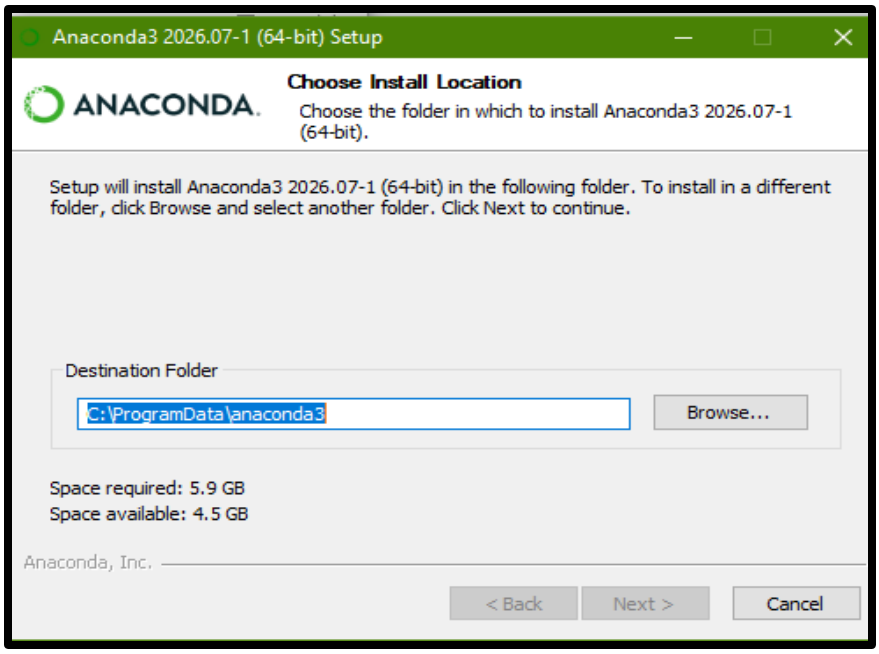
Step 3:



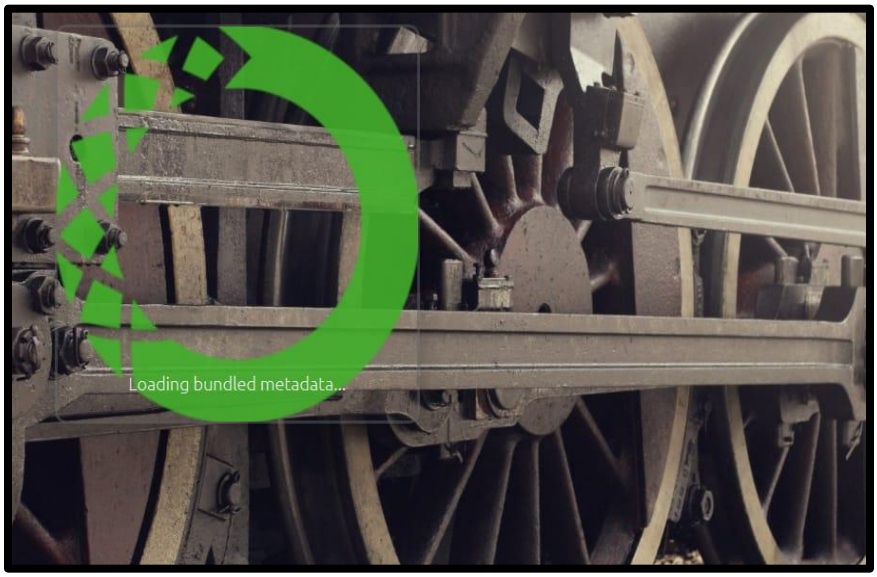
Step 4:



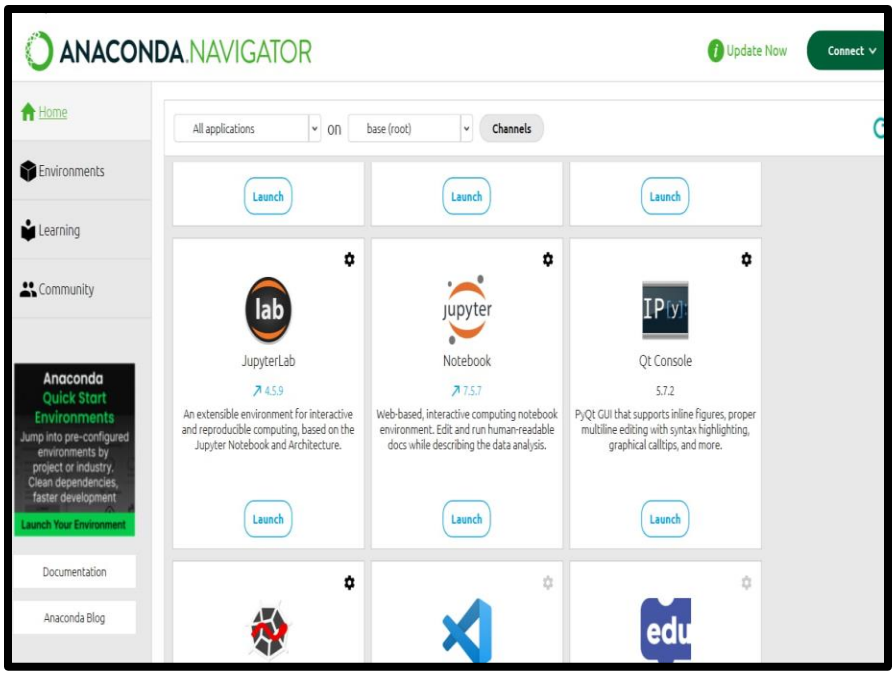
Step 5:


Anaconda3 2026.07-1 (64-bit) Setup
ANACONDA.
Choose Install Location
Choose the folder in which to install Anaconda3 2026.07-1 (64-bit).
Setup will install Anaconda3 2026.07-1 (64-bit) in the following folder. To install in a different folder, click Browse and select another folder. Click Next to continue.
Destination Folder
C:\ProgramData\anaconda3
Browse...
Space required: 5.9 GB
Space available: 4.5 GB
Anaconda, Inc.
< Back
Next >
Cancel

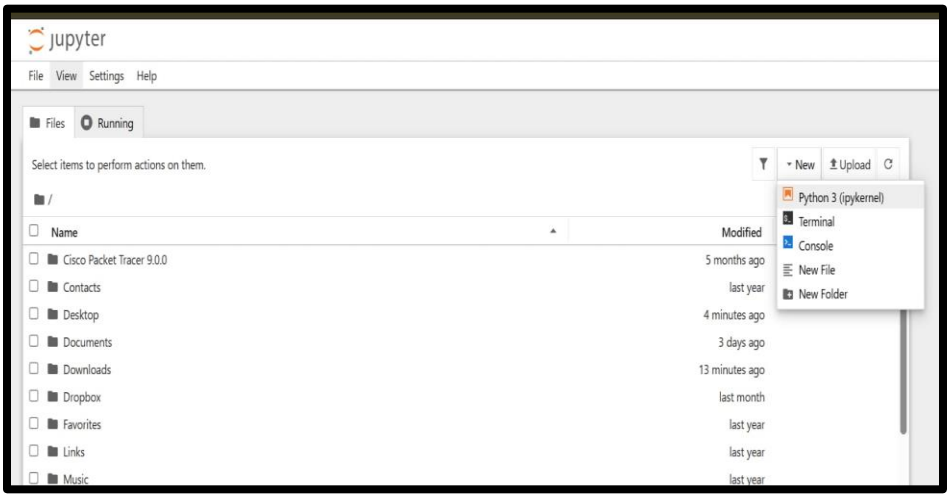
Step 6:


Loading bundled metadata...

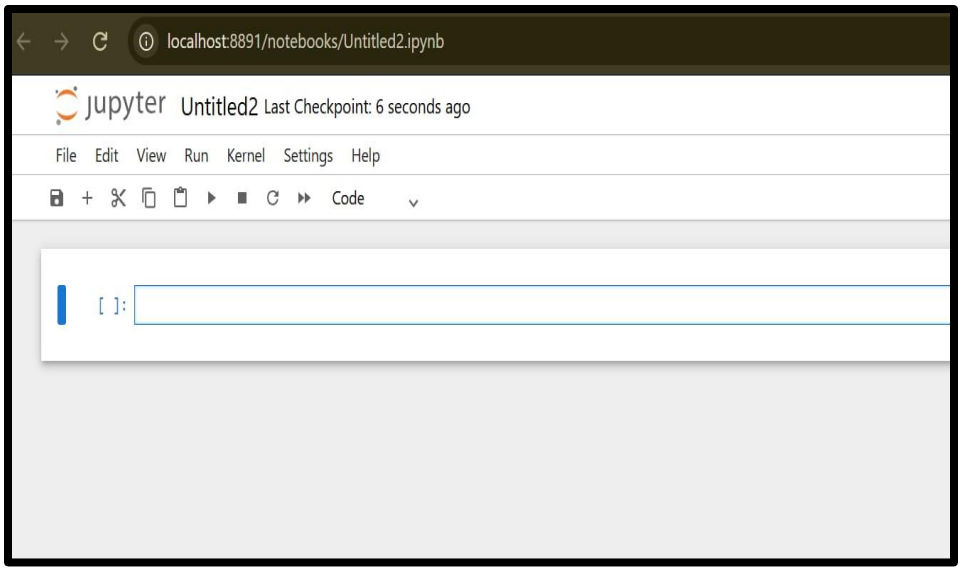
Step 7:


ANACONDA.NAVIGATOR
Update Now
Connect
Home
Environments
Learning
Community
All applications
on
base (root)
Channels
Launch
JupyterLab
4.5.9
An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.
Notebook
7.5.7
Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.
Qt Console
5.7.2
PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.
Anaconda
Quick Start
Environments
Jump into pre-configured environments by project or industry. Clean dependencies, faster development
Launch Your Environment
Documentation
Anaconda Blog

Step 8:

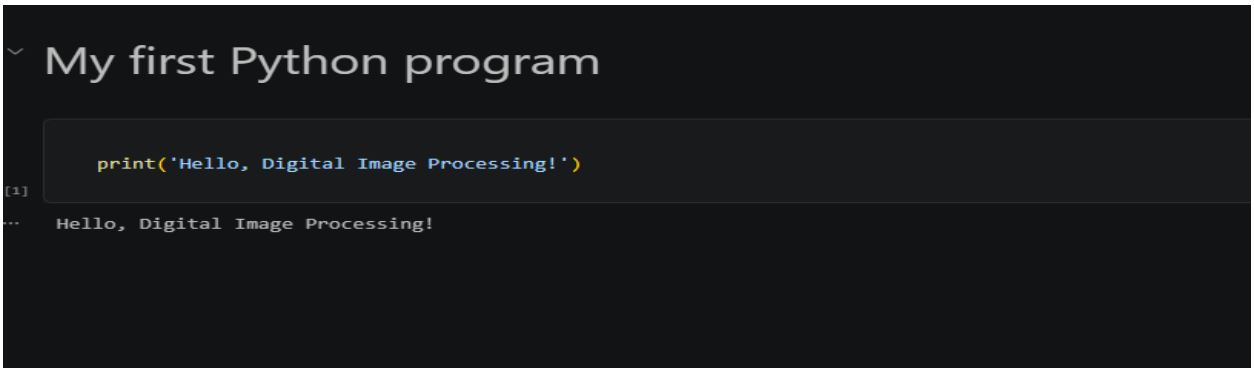

jupyter
File
View
Settings
Help
Files
Running
Select items to perform actions on them.
New
Upload
Python 3 (ipykernel)
Terminal
Console
New File
New Folder
Name
Modified
Cisco Packet Tracer 9.0.0
5 months ago
Contacts
last year
Desktop
4 minutes ago
Documents
3 days ago
Downloads
13 minutes ago
Dropbox
last month
Favorites
last year
Links
last year
Music
last year

Step 9:



Task 2: Create three Python files:

File name	Purpose
hello.py	Run your first Python program
basics.py	Solve all six exercises
lab01_imports.py	Import the required libraries



```
import sys

print("Python version:", sys.version)
print("Python executable:", sys.executable)
```

```
Python version: 3.14.6 | packaged by Anaconda, Inc. | (main, Jul  9 2026, 14:29:05) [MSC v.1942 64 bit (AMD64)]
Python executable: D:\Anaconda3\python.exe
```

```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt

print("All three packages imported successfully!")
```

```
All three packages imported successfully!
```

➢ Part 2

Exercise 1 — Variables and data types

Question: Create variables for image width, height, name, and color status. Print their values and types, then calculate the total pixels.

Solution:

Exercise 01: Variables And Datatypes

```
width = 640
height = 480
image_name = "sample.png"
is_color = True

print(width, type(width))
print(height, type(height))
print(image_name, type(image_name))
print(is_color, type(is_color))

total_pixels = width * height
print("Total pixels:", total_pixels)
```

```
640 <class 'int'>
480 <class 'int'>
sample.png <class 'str'>
True <class 'bool'>
Total pixels: 307200
```

Explanation

- width = 640 stores the image width as an integer (int).
- height = 480 stores the image height as an integer.
- image_name = "sample.png" stores text, called a string (str).
- is_color = True stores a Boolean value (bool), meaning true or false.
- type() tells you the data type of a value.
- width * height multiplies the width by the height.

Exercise 2 — if, elif, and else

Question: Label an intensity as Invalid, Bright pixel, or Dark pixel.
An intensity below 0 or above 255 is invalid. A valid intensity of 128 or more is bright; otherwise, it is dark.

Solution:



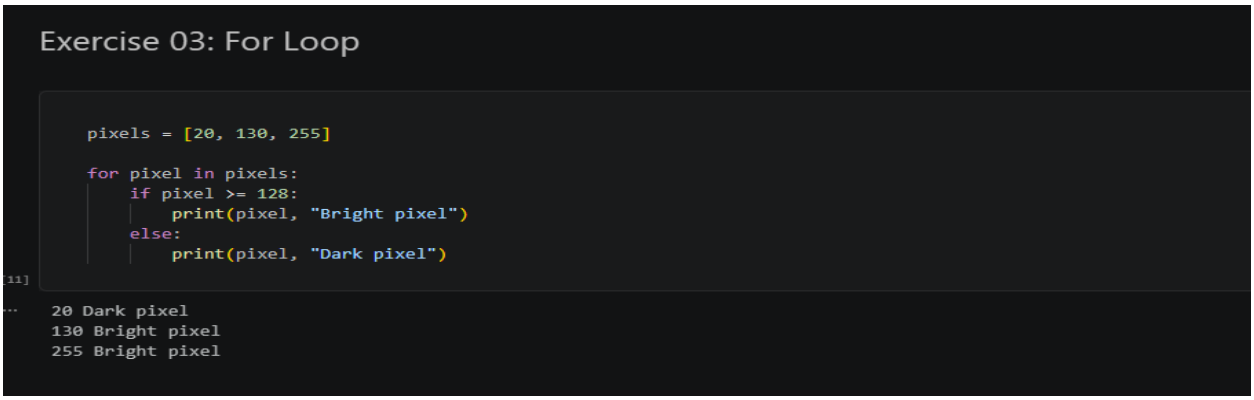
Explanation

- if checks the first condition.
- or means either condition can be true.

- elif checks another condition if the first one is false.
- else handles the remaining cases.
- The for loop repeats the checks for every number in the list.

Exercise 3 — for and while loops

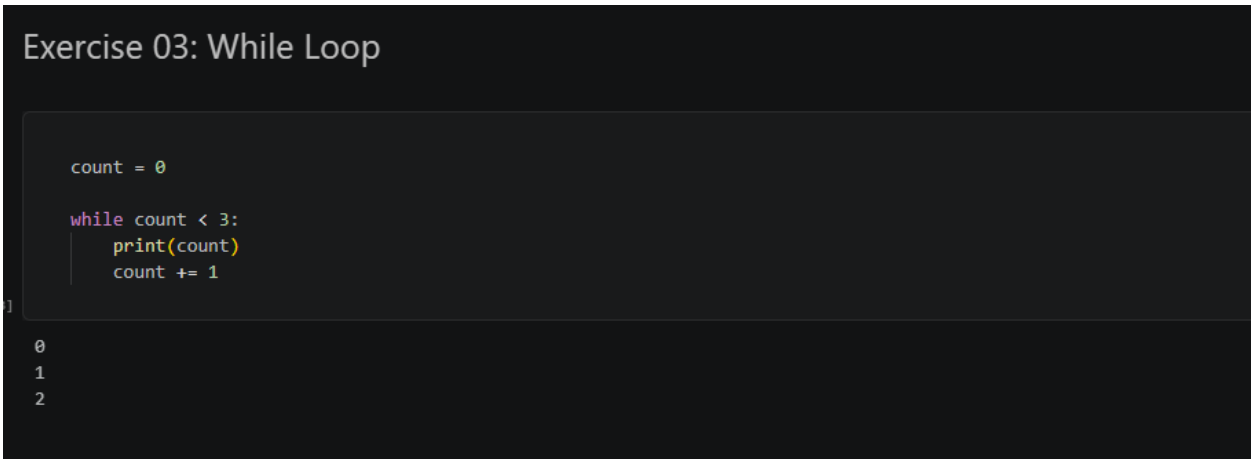
Question: Use a for loop to label pixel values [20, 130, 255], then use a while loop to print 0, 1, and 2.



Explanation

The for loop: It visits each item in the list one by one.

- 20 is a Dark pixel.
- 130 is a Bright pixel.
- 255 is a Bright pixel.



The while loop: It keeps running while count < 3 is true.

1. Initially, count = 0.
2. It prints 0, then increases the count to 1.
3. It prints 1, then increases the count to 2.
4. It prints 2, then increases the count to 3.
5. The condition becomes false, so the loop stops.

Exercise 4 — Functions

Question: Create a function called is_bright() that returns True if the pixel intensity is at least the threshold and False otherwise**.**

Exercise: 04

```
def is_bright(pixel, threshold=128):
    return pixel >= threshold

print(is_bright(127))
print(is_bright(128))
print(is_bright(200, 210))
```

```
False
True
False
```

Explanation

- def is used to define a function.
- pixel is the value being checked.
- threshold=128 is the default threshold.
- return sends the result back to the place where the function was called.

Exercise 5 — Lists and tuples

Question: Modify a list, print its length, and unpack a tuple into width and height.

Exercise 05: Lists And Tuples

```
pixels = [0, 128, 255]

print("First item:", pixels[0])

pixels[1] = 150
pixels.append(64)

print("Updated list:", pixels)
print("Length:", len(pixels))

size = (640, 480)
width, height = size

print("Width:", width)
print("Height:", height)

# A list is mutable, so its items can be changed.
# A tuple is immutable, so its items cannot be reassigned.
```

```
First item: 0
Updated list: [0, 150, 255, 64]
Length: 4
Width: 640
```

Do you want to install the recommended 'Python from Microsoft for the Python language?

Install

Explanation

Lists store multiple values and can be changed.

- pixels[0] accesses the first item. Python indexing starts at 0.
- pixels[1] = 150 replaces the second item.
- append(64) adds a new item to the end.
- len(pixels) counts the items.

Tuples also store multiple values, but their items cannot be changed after creation.

Exercise 6 — Dictionaries and sets

Question: Add a mode to an image dictionary, access its values using keys, and count unique labels with a set.



Explanation

Dictionary: Stores data as key-value pairs.

In this example:

- "width" is a key with value 640.

- "height" is a key with value 480.
- image["mode"] = "RGB" adds a new key-value pair.
- image["width"] retrieves the value associated with the width key.

➢ Part 3

Explaining Python versus pip and the purpose of the import statements.

Python is a programming language used to write and run programs, while pip is a package installer used to install additional Python libraries such as NumPy, Pillow, and Matplotlib. Import statements allow us to use the functions and tools provided by these libraries in our Python programs. For example, import numpy as np allows us to use NumPy for numerical calculations and arrays.

*****THE END*****